Supplementary Material: Zero-Incoherence Capacity of Interactive Encoding Systems

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
AFM1	FactMatroid.determinesFact_iff_mem_factSpan Ssot/FactMatroid.lean	AFM2	FactMatroid.factMatroid_indep_iff Ssot/FactMatroid.lean
AFM3	FactMatroid.basisFacts_card_eq_finrank Ssot/FactMatroid.lean	AFM4	FactMatroid.basisFacts_minimal Ssot/FactMatroid.lean
AFM5	FactMatroid.minimalDetermining_basis Ssot/FactMatroid.lean	AFM6	FactMatroid.factRankFinset_le_card Ssot/FactMatroid.lean
AFM7	FactMatroid.coordProjection_range_le_card Ssot/FactMatroid.lean	AFM8	FactMatroid.factRankFinset_eq_card_of_ indepFacts Ssot/FactMatroid.lean
AFM9	FactMatroid.factRankFinset_eq_total_of_ determinesAllFinset Ssot/FactMatroid.lean	AFM10	FactMatroid.factSpanFinset_eq_range_ coordProjection_dualMap Ssot/FactMatroid.lean
AFM11	FactMatroid.factRankFinset_eq_finrank_range_ coordProjection Ssot/FactMatroid.lean	AFM12	FactMatroid.factRankFinset_le_finrank_ directionSpace Ssot/FactMatroid.lean
AFM13	FactMatroid.coordProjection_range_eq_card_of_ indepFacts Ssot/FactMatroid.lean	AFM14	FactMatroid.coordProjection_range_eq_total_of_ determinesAllFinset Ssot/FactMatroid.lean
BND1	ssot_upper_bound Ssot/Bounds.lean	BND2	non_ssot_lower_bound Ssot/Bounds.lean
BND3	ssot_advantage_unbounded Ssot/Bounds.lean	CAP2	ssot_guarantees_coherence Ssot/Coherence.lean
CAP3	non_ssot_permits_incoherence Ssot/Coherence.lean	CC1	CC.derivation_preserves_coherence_core Ssot/ClaimClosure.lean
CIA1	ClaimClosure.side_information_requirement Ssot/ClaimClosure.lean	CIA3	ClaimClosure.finite_counting_converse Ssot/ClaimClosure.lean
CIA4	ClaimClosure.info_dof_counting_contradiction Ssot/ClaimClosure.lean	COH1	dof_one_implies_coherent Ssot/Coherence.lean
COH2	dof_gt_one_incoherence_possible Ssot/Coherence.lean	CRG1	Deps.cargo_ssot_incomplete Ssot/LangDeps.lean
CRG2	Deps.cargo_has_introspection Ssot/LangDeps.lean	CRG3	Deps.cargo_no_hooks Ssot/LangDeps.lean
DBE1	Database.external_etl_ssot_incomplete Ssot/LangDatabase.lean	DBE2	Database.external_etl_no_hooks Ssot/LangDatabase.lean
DBE3	Database.external_etl_no_introspection Ssot/LangDatabase.lean	DBV1	Database.engine_view_ssot_complete Ssot/LangDatabase.lean
DBV2	Database.engine_view_has_hooks Ssot/LangDatabase.lean	DBV3	Database.engine_view_has_introspection Ssot/LangDatabase.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
DER2	ClaimClosure.derivation_preserves_coherence_ core Ssot/ClaimClosure.lean	FM6	FM.factRankFinset_eq_card_of_indepFacts Ssot/FactMatroid.lean
GOL1	StaticLang.go_no_inheritance_hooks Ssot/LangStatic.lean	IND1	both_requirements_independent Ssot/Requirements.lean
IND2	both_requirements_independent' Ssot/Requirements.lean	JVA1	StaticLang.java_lacks_definition_hooks Ssot/LangStatic.lean
MFT3	MultiFact.exactRecovery_iff_properColoring Ssot/MultiFact.lean	MFT4	MultiFact.exists_exactRecovery_iff_colorable Ssot/MultiFact.lean
MFT5	MultiFact.binaryViews_nonclique_witness Ssot/MultiFact.lean	MFT6	MultiFact.binaryViews_colorable_two Ssot/MultiFact.lean
MFT7	MultiFact.binaryViews_not_colorable_one Ssot/MultiFact.lean	MFT8	MultiFact.binaryViews_oneTag_uniform_ successProb_le_half Ssot/MultiFact.lean
MFT9	MultiFact.exists_exactOn_iff_colorableOn Ssot/MultiFact.lean	MFT10	MultiFact.binaryViews_oneTag_exactOn_card_le_ two Ssot/MultiFact.lean
MFT11	MultiFact.pairProperColoring_of_ componentColorings Ssot/MultiFact.lean	MFT12	MultiFact.pairColorable_of_colorable Ssot/MultiFact.lean
MFT13	MultiFact.pair_product_clique_budget_lower_ bound Ssot/MultiFact.lean	MFT14	MultiFact.exists_exactOn_card_eq_ maxColorableCard Ssot/MultiFact.lean
MFT15	MultiFact.binaryViews_maxColorableCard_one_eq_ two Ssot/MultiFact.lean	MFT16	MultiFact.pair_product_success_card_lower_ bound Ssot/MultiFact.lean
MFT17	MultiFact.pair_product_independent_lower_bound Ssot/MultiFact.lean	MFT18	MultiFact.exactOn_mass_le_maxColorableMass Ssot/MultiFact.lean
MFT19	MultiFact.exists_exactOn_mass_eq_ maxColorableMass Ssot/MultiFact.lean	MFT20	MultiFact.colorable_iff_graphColorable Ssot/MultiFact.lean
MFT21	MultiFact.block_power_independent_lower_bound Ssot/MultiFact.lean	MFT22	MultiFact.block_add_independent_lower_bound Ssot/MultiFact.lean
MFT23	MultiFact.exactRateDistortionValue_upper Ssot/MultiFact.lean	MFT24	MultiFact.exists_exactRateDistortionValue Ssot/MultiFact.lean
MFT25	MultiFact.pairConfusable_iff_strongProdAdj Ssot/MultiFact.lean	MFT26	MultiFact.pairConfusabilityGraph_eq_strongProd Ssot/MultiFact.lean
MFT29	MultiFact.concatBlockConfusable_iff_ strongProdAdj Ssot/MultiFact.lean	MFT30	MultiFact.blockConfusabilityGraph_ addIsoStrongProd Ssot/MultiFact.lean
MFT31	MultiFact.blockMaxIndependentCard_eq_ graphMaxIndependentCard Ssot/MultiFact.lean	MFT32	MultiFact.shannonLowerCapacity_eq_iSup_ graphRates Ssot/MultiFact.lean
MFT33	MultiFact.blockRate_le_blockRate_mul Ssot/MultiFact.lean	MFT34	MultiFact.blockConfusable_iff_strongPowAdj Ssot/MultiFact.lean
MFT35	MultiFact.blockConfusabilityGraph_eq_strongPow Ssot/MultiFact.lean	MFT36	MultiFact.shannonLowerCapacity_eq_ graphShannonLowerCapacity Ssot/MultiFact.lean
MFT38	MultiFact.strongPow_addIsoStrongProd Ssot/MultiFact.lean	MFT39	MultiFact.graphMaxIndependentCard_strongPow_ add_eq_strongProd Ssot/MultiFact.lean
MFT43	MultiFact.graphNegLogSeq_subadditive Ssot/MultiFact.lean	MFT44	MultiFact.tendsto_graphPowerRateNat_ graphShannonCapacityReal Ssot/MultiFact.lean
MFT45	MultiFact.graphPowerRate_le_ graphShannonCapacityReal Ssot/MultiFact.lean	MFT46	MultiFact.graphShannonCapacityReal_nonneg Ssot/MultiFact.lean
MFT47	MultiFact.iSup_graphPowerRate_eq_ graphShannonCapacityReal Ssot/MultiFact.lean	MFT48	MultiFact.shannonCapacityReal_eq_iSup_ blockRate Ssot/MultiFact.lean
MFT49	MultiFact.blockRate_le_shannonCapacityReal Ssot/MultiFact.lean	MFT50	MultiFact.compl_strongPow_adj_iff_exists Ssot/MultiFact.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
MFT51	MultiFact.complStrongPow_colorable Ssot/MultiFact.lean	MFT52	MultiFact.graphMaxIndependentCard_strongPow_ le_complChromaticPow Ssot/MultiFact.lean
MFT53	MultiFact.graphPowerRate_le_log_complChromatic Ssot/MultiFact.lean	MFT54	MultiFact.graphShannonCapacityReal_le_log_ complChromatic Ssot/MultiFact.lean
MFT55	MultiFact.graphShannonCapacityReal_le_log_ complChromaticNumber Ssot/MultiFact.lean	MFT56	MultiFact.graphShannonLowerCapacity_eq_ofReal_ graphShannonCapacityReal Ssot/MultiFact.lean
MFT57	MultiFact.graphThetaPSDLogUpper_eq_ graphThetaLogUpper Ssot/MultiFact.lean	MFT58	MultiFact.graphShannonCapacityReal_le_ graphThetaPSDLogUpper Ssot/MultiFact.lean
MFT59	MultiFact.graphThetaPSDLogUpper_strongProd_le Ssot/MultiFact.lean	MFT60	MultiFact.graphThetaPSDLogSeq_subadditive Ssot/MultiFact.lean
MFT61	MultiFact.tendsto_graphThetaPSDPowerRateNat_ graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean	MFT62	MultiFact.graphThetaPSDAsymptoticUpper_le_ graphThetaPSDLogUpper Ssot/MultiFact.lean
MFT63	MultiFact.iInf_graphThetaPSDPowerRate_eq_ graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean	MFT64	MultiFact.lovaszOrthoLogUpper_eq_ graphThetaLogUpper Ssot/MultiFact.lean
MFT65	MultiFact.lovaszThetaPrimalLogUpper_eq_ graphThetaPSDLogUpper Ssot/MultiFact.lean	MFT66	MultiFact.graphThetaLiftedDualLogUpper_eq_ graphThetaSchurDualLogUpper Ssot/MultiFact.lean
MFT67	MultiFact.graphOneShotLogMaxIndependentCard_ le_graphThetaLogUpper Ssot/MultiFact.lean	MFT68	MultiFact.graphPowerRate_le_ graphThetaPSDPowerRate Ssot/MultiFact.lean
MFT69	MultiFact.graphShannonCapacityReal_le_ graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean	MFT70	MultiFact.lovaszThetaPrimalAsymptoticUpper_eq_ graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean
MFT71	MultiFact.graphShannonCapacityReal_le_ lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean	MFT72	MultiFact.graphThetaSchurDualLogUpper_ strongProd_le Ssot/MultiFact.lean
MFT73	MultiFact.graphThetaSchurDualLogUpper_iso_eq Ssot/MultiFact.lean	MFT74	MultiFact.graphThetaSchurDualLogSeq_ subadditive Ssot/MultiFact.lean
MFT75	MultiFact.tendsto_graphThetaSchurDualPowerRateNat_ graphThetaSchurDualAsymptoticUpper Ssot/MultiFact.lean	MFT76	MultiFact.iInf_graphThetaSchurDualPowerRate_ eq_graphThetaSchurDualAsymptoticUpper Ssot/MultiFact.lean
MFT77	MultiFact.graphThetaSchurDualAsymptoticUpper_ le_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean	MFT78	MultiFact.graphThetaSchurDualAsymptoticUpper_ le_lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean
MFT79	MultiFact.lovaszThetaDualLogUpper_eq_ graphThetaSchurDualLogUpper Ssot/MultiFact.lean	MFT80	MultiFact.lovaszThetaDualAsymptoticUpper_eq_ graphThetaSchurDualAsymptoticUpper Ssot/MultiFact.lean
MFT81	MultiFact.lovaszThetaDualAsymptoticUpper_le_ lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean	MFT82	MultiFact.lovaszThetaDualLogUpper_eq_ lovaszThetaPrimalLogUpper Ssot/MultiFact.lean
MFT83	MultiFact.lovaszThetaDualAsymptoticUpper_eq_ lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean	MFT84	MultiFact.graphShannonCapacityReal_le_ lovaszThetaAsymptoticUpper Ssot/MultiFact.lean
MFT85	MultiFact.graphShannonCapacityReal_eq_ lovaszThetaAsymptoticUpper_clusterGraph Ssot/MultiFact.lean	MFT86	MultiFact.graphShannonCapacityReal_eq_log_ card_clusterGraph Ssot/MultiFact.lean
MFT87	MultiFact.lovaszThetaAsymptoticUpper_eq_log_ card_clusterGraph Ssot/MultiFact.lean	MFT88	MultiFact.confusabilityGraph_eq_clusterGraph_ transcriptLabel Ssot/MultiFact.lean
MFT89	MultiFact.shannonCapacityReal_eq_ shannonLovaszThetaAsymptoticUpper_of_ fiberCoherent Ssot/MultiFact.lean	MFT90	MultiFact.shannonCapacityReal_eq_log_ transcriptFiberCard_of_fiberCoherent Ssot/MultiFact.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
MFT91	MultiFact.shannonLovaszThetaAsymptoticUpper_eq_log_transcriptFiberCard_of_fiberCoherent Ssot/MultiFact.lean	MFT96	MultiFact.shannonLovaszThetaAsymptoticUpper_eq_log_connectedComponents_of_confusableTransitive Ssot/MultiFact.lean
MFT100	MultiFact.shannonCapacityReal_eq_shannonLovaszThetaAsymptoticUpper_of_meetWitnessed Ssot/MultiFact.lean	MFT101	MultiFact.shannonCapacityReal_eq_log_connectedComponents_of_meetWitnessed Ssot/MultiFact.lean
MFT102	MultiFact.shannonLovaszThetaAsymptoticUpper_eq_log_connectedComponents_of_meetWitnessed Ssot/MultiFact.lean	MFT107	MultiFact.ViewCompositionClosed Ssot/MultiFact.lean
MFT108	MultiFact.confusableTransitive_iff_viewCompositionClosed Ssot/MultiFact.lean	MFT109	MultiFact.confusable_iff_exists_viewClusterAdj Ssot/MultiFact.lean
MFT110	MultiFact.SupportConfusable Ssot/MultiFact.lean	MFT111	MultiFact.deterministicSupportConfusable_iff_confusable Ssot/MultiFact.lean
MFT112	MultiFact.deterministicSupportConfusabilityGraph_eq_confusabilityGraph Ssot/MultiFact.lean	MFT113	MultiFact.card_state_eq Ssot/MultiFact.lean
MFT114	MultiFact.confusableOracle_eq_true_iff Ssot/MultiFact.lean	MFT115	MultiFact.confusableOrderedPairs Ssot/MultiFact.lean
MFT116	MultiFact.card_state_prod_eq Ssot/MultiFact.lean	MFT117	MultiFact.card_confusableOrderedPairs_le_naive_bound Ssot/MultiFact.lean
MFT118	MultiFact.observe_eq_iff_subset_agreeSet Ssot/MultiFact.lean	MFT119	MultiFact.confusable_iff_exists_view_subset_agreeSet Ssot/MultiFact.lean
MFT120	MultiFact.agreeSet_statePerm_eq Ssot/MultiFact.lean	MFT121	MultiFact.confusable_statePerm_iff Ssot/MultiFact.lean
MFT122	MultiFact.confusable_congr_agreeSet Ssot/MultiFact.lean	MFT123	MultiFact.confusabilityGraph_statePermIso Ssot/MultiFact.lean
MFT124	MultiFact.exists_confusabilityGraph_iso_send Ssot/MultiFact.lean	MFT125	MultiFact.agreementUpwardFamily_upwardClosed Ssot/MultiFact.lean
MFT126	MultiFact.confusable_iff_agreeSet_mem_agreementUpwardFamily Ssot/MultiFact.lean	MFT127	MultiFact.confusable_viewFamilyOfAgreementUpwardFamily_iff Ssot/MultiFact.lean
MFT128	MultiFact.exists_viewFamily_iff_agreementClassified Ssot/MultiFact.lean	MFT129	MultiFact.agreeSet_eq_univ_iff Ssot/MultiFact.lean
MFT130	MultiFact.exists_distinct_states_with_agreeSet_eq_iff Ssot/MultiFact.lean	MFT131	MultiFact.upwardClosed_family_eq_on_proper_subsets_of_same_relation Ssot/MultiFact.lean
MFT132	MultiFact.agreementOracle_eq_true_iff Ssot/MultiFact.lean	MFT133	MultiFact.agreementOracle_eq_confusableOracle Ssot/MultiFact.lean
MFT134	MultiFact.agreementOracleViewWork_le Ssot/MultiFact.lean	MFT135	MultiFact.agreementOracleTotalWork_le Ssot/MultiFact.lean
MFT136	MultiFact.meetWitnessed_implies_transitive_confusability Ssot/MultiFact.lean	MFT137	MultiFact.transitive_confusability_implies_cluster_graph Ssot/MultiFact.lean
NPM1	Deps.npm_ssot_incomplete Ssot/LangDeps.lean	NPM2	Deps.npm_has_introspection Ssot/LangDeps.lean
NPM3	Deps.npm_no_hooks Ssot/LangDeps.lean	OBS1	ObserverModel.exact_recovery_implies_pair_injective paper1/Paper1IT/ObserverTagModel.lean
OBS2	ObserverModel.pair_injective_implies_exact_recovery paper1/Paper1IT/ObserverTagModel.lean	OBS3	ObserverModel.fiber_card_le_tag_alphabet paper1/Paper1IT/ObserverTagModel.lean
OBS4	ObserverModel.exact_recovery_global_count paper1/Paper1IT/ObserverTagModel.lean	OBS5	ObserverModel.clique_card_le_tag_alphabet paper1/Paper1IT/ObserverTagModel.lean
PNM1	Deps.pnpm_ssot_incomplete Ssot/LangDeps.lean	PNM2	Deps.pnpm_has_introspection Ssot/LangDeps.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
PNM3	Deps.pnpm_no_hooks Ssot/LangDeps.lean	POT1	Deps.poetry_ssot_incomplete Ssot/LangDeps.lean
POT2	Deps.poetry_has_introspection Ssot/LangDeps.lean	POT3	Deps.poetry_no_hooks Ssot/LangDeps.lean
PRB47	ObserverModel.sourceEntropy_ eq_observationTagEntropy_add_ conditionalEntropyGivenPair paper1/Paper1IT/ProbabilisticFinite.lean	PRB55	ObserverModel.successSet_clique_entropy_le_ log_tags paper1/Paper1IT/FanoFinite.lean
PRB68	ObserverModel.deterministicKernel_entropy_ data_processing paper1/Paper1IT/ProbabilisticFinite.lean	PRB73	ObserverModel.observationTagEntropy_gap_nonneg paper1/Paper1IT/ProbabilisticFinite.lean
PRB74	ObserverModel.observationTagEntropy_gap_eq_ zero_of_klDiv_zero_uniform paper1/Paper1IT/ProbabilisticFinite.lean	PRB81	ObserverModel.decodedOutputEntropy_le_ mutualInfoDeterministic paper1/Paper1IT/ProbabilisticFinite.lean
PRB82	ObserverModel.decodedOutputEntropy_source_gap_ nonneg paper1/Paper1IT/ProbabilisticFinite.lean	PRB83	ObserverModel.decodedOutputEntropy_log_gap_ nonneg paper1/Paper1IT/ProbabilisticFinite.lean
PRB85	ObserverModel.decodedOutputEntropy_fano_ budgeted paper1/Paper1IT/FanoFinite.lean	PRB87	ObserverModel.decodedOutputEntropy_fano_ observation_only paper1/Paper1IT/FanoFinite.lean
PRB90	ObserverModel.DeterministicObservable.entropy_ coarsen_le_sourceEntropy paper1/Paper1IT/ProbabilisticFinite.lean	PRB93	ObserverModel.decodedOutputEntropy_gap_eq_ zero_of_klDiv_zero_uniform paper1/Paper1IT/ProbabilisticFinite.lean
PRB94	ObserverModel.decodedOutputEntropy_gap_pos_ implies_klDiv_ne_zero paper1/Paper1IT/ProbabilisticFinite.lean	PRB95	ObserverModel.decodedOutputObservable_entropy_ le_sourceEntropy paper1/Paper1IT/ProbabilisticFinite.lean
PYH1	Python.python_has_hooks Ssot/LangPython.lean	PYI1	Python.python_has_introspection Ssot/LangPython.lean
RAT1	ClaimClosure.rate_incoherence_step Ssot/ClaimClosure.lean	RED1	ClaimClosure.redundancy_incoherence_equiv Ssot/ClaimClosure.lean
REQ1	structural_facts_fixed_at_definition Ssot/Foundations.lean	REQ2	definition_hooks_necessary Ssot/Requirements.lean
REQ3	introspection_necessary_for_verification Ssot/Requirements.lean	RUH1	Rust.rust_has_definition_hooks Ssot/LangRust.lean
RUI1	Rust.rust_lacks_introspection Ssot/LangRust.lean	SOT1	ssot_iff Ssot/Completeness.lean
TRI1	timing_trichotomy_exhaustive Ssot/Foundations.lean	TSE1	StaticLang.ts_types_erased Ssot/LangStatic.lean
TSN1	StaticLang.ts_no_runtime_type_definitions Ssot/LangStatic.lean		

2 Claim Mapping to Lean Handles

This section provides the direct mapping between paper claims and the mechanized proof handles in the Lean 4 artifact.

Paper claim	**Lean handle**
Proposition VIII.3: View-fiber clique sizes	`FM6`, `AFM7`, `AFM8`, `AFM11`, `AFM9`, `AFM6`, `AFM12`, `AFM10`
Definition XI.2: Verifiable Structural Integrity	`OBS5`
Lemma X.4: Information-Constrained DOF Lower Bound	`CIA4`
Proposition VIII.1: Affine fact-matroid specialization	`AFM3`, `AFM4`, `AFM1`, `AFM2`, `AFM5`
Theorem V.1: Asymptotic Shannon-capacity theorem	`MFT49`, `MFT43`, `MFT45`, `MFT46`, `MFT56`, `MFT47`, `MFT48`, `MFT44`
Definition XI.6: Zero-Delay State Synchronization	`REQ2`
Definition II.7: Independent Location	`RAT1`, `CAP3`, `CAP2`
Theorem III.2: Confusability formulation	`OBS5`
Definition IX.2: Derivation	`DER2`
Corollary III.3: Counting converse	`CIA3`
Proposition III.4: Budgeted finite-error converse	`PRB85`, `PRB87`
Theorem XI.9: Requirements are Independent	`IND1`, `IND2`
Theorem X.3: Above-Threshold Lower Bound	`BND2`
Theorem IV.1: Exact recovery as graph colorability	`MFT20`, `MFT3`, `MFT4`
Theorem IV.3: Exact finite-error success budget	`MFT15`, `MFT14`
Theorem IV.4: First non-clique witness	`MFT6`, `MFT5`, `MFT7`, `MFT10`, `MFT8`
Theorem IV.5: Block-composition law	`MFT12`, `MFT11`, `MFT13`, `MFT16`
Theorem IV.2: Success-set characterization	`MFT9`
Theorem III.1: Pair-injectivity characterization	`OBS1`, `OBS2`
Definition XI.8: Structural Side-Information	`REQ3`
Corollary XI.10: Operational Realizability Criterion	`SOT1`
Definition XI.4: Structural Fact	`REQ1`, `TRI1`
Theorem X.5: Unbounded Gap	`BND3`
Definition X.2: Modification Cost Model	`BND1`

Auto summary: mapped 24/24 (full=24, derived=0, unmapped=0).